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const express=require('express'),multer=require('multer'),bcrypt=require('bcryptjs')
const PORT=process.env.PORT||3000,DAY=864e5;
const DATA=path.join(__dirname,'data'),UP=path.join(__dirname,'uploads'),DB=path.
fs.mkdirSync(DATA,{recursive:true});fs.mkdirSync(UP,{recursive:true});
let db=fs.existsSync(DB)?JSON.parse(fs.readFileSync(DB)):{admin:null,users:[],vid
const save=()=>fs.writeFileSync(DB,JSON.stringify(db,null,2));
const rid=()=>crypto.randomBytes(6).toString('hex');
if(!db.admin){const l=process.env.ADMIN_LOGIN||'admin',p=process.env.ADMIN_PASSWO
const upload=multer({storage:multer.diskStorage({destination:UP,filename:(q,f,cb)
const app=express();app.use(express.json(),cookieParser());
const who=q=>{const s=q.cookies.sid;if(!s)return null;if(db.admin.sid===s)return{
const needUser=(q,r,n)=>{const w=who(q);if(!w)return r.status(401).json({error:'A
const needAdmin=(q,r,n)=>{const w=who(q);if(!w||w.role!=='admin')return r.status(
const tries={};
app.post('/api/login',(q,r)=>{
  const t=tries[q.ip]||{n:0,at:0};
  if(t.n>=5&&Date.now()-t.at<9e5)return r.status(429).json({error:'Тым көп әрекет
  const{login='',password=''}=q.body;let role=null,rec=null;
  if(login===db.admin.login&&bcrypt.compareSync(password,db.admin.hash)){role='ad
  else{const u=db.users.find(u=>u.login===login);if(u&&bcrypt.compareSync(passwor
  if(!rec){tries[q.ip]={n:t.n+1,at:Date.now()};return r.status(401).json({error:'
  delete tries[q.ip];
  if(role==='user'&&rec.expires<=Date.now())return r.status(403).json({error:'Док
  rec.sid=crypto.randomBytes(24).toString('hex');save();
  r.cookie('sid',rec.sid,{httpOnly:true,sameSite:'lax',maxAge:60*DAY});r.json({ro
});
app.post('/api/logout',(q,r)=>{r.clearCookie('sid');r.json({ok:1})});
app.get('/api/me',needUser,(q,r)=>r.json(q.w.role==='admin'?{role:'admin'}:{role:
app.get('/api/videos',needUser,(q,r)=>r.json(db.videos.map(({id,title})=>({id,tit
app.get('/api/video/:id',needUser,(q,r)=>{const v=db.videos.find(v=>v.id===q.para
// админ
app.get('/api/admin/users',needAdmin,(q,r)=>r.json(db.users.map(({id,login,expire
app.post('/api/admin/users',needAdmin,(q,r)=>{
  const{login,password,days}=q.body;
  if(!login||!password||password.length<6)return r.status(400).json({error:'Логин
  if(![14,30,60].includes(+days))return r.status(400).json({error:'Мерзім: 14, 30
  if(db.users.some(u=>u.login===login)||login===db.admin.login)return r.status(40
  db.users.push({id:rid(),login,hash:bcrypt.hashSync(password,10),expires:Date.no
});
app.post('/api/admin/users/:id/extend',needAdmin,(q,r)=>{
  const u=db.users.find(u=>u.id===q.params.id);const d=q.body.days;
  if(!u||![14,30,60].includes(d))return r.sendStatus(400);
  u.expires=Math.max(Date.now(),u.expires)+d*DAY;save();r.json({ok:1});
});

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app.post('/api/admin/users/:id/revoke',needAdmin,(q,r)=>{const u=db.users.find(u=
app.delete('/api/admin/users/:id',needAdmin,(q,r)=>{db.users=db.users.filter(u=>u
app.post('/api/admin/videos',needAdmin,upload.single('video'),(q,r)=>{
  if(!q.file)return r.status(400).json({error:'Видео файлын таңдаңыз'});
  db.videos.push({id:rid(),title:(q.body.title||q.file.originalname).slice(0,120)
});
app.delete('/api/admin/videos/:id',needAdmin,(q,r)=>{
  const v=db.videos.find(v=>v.id===q.params.id);if(v){fs.rmSync(path.join(UP,v.fi
});
// ===== ТЕСТ МОДУЛИ (қазақша + орысша) =====
db.questions=db.questions||[];db.topics=db.topics||Array.from({length:31},(_,i)=>
db.questions.forEach(x=>{if(!x.kk&&x.text){x.kk={text:x.text,options:x.options,ex
const lg=l=>l==='ru'?'ru':'kk',attempts={};
const shuf=a=>{a=[...a];for(let i=a.length-1;i>0;i--){const j=crypto.randomInt(i+
const tc=(l,n)=>db.questions.filter(x=>x[l]&&!x.review&&(!n||x.topic===n)).length
const ttl=(t,l)=>{const p=t.split('|').map(s=>s.trim());return(l==='ru'?p[1]:p[0]
app.get('/api/test/info',needUser,(q,r)=>{const l=lg(q.query.lang);r.json({total:
  wrong:(q.w.u?.wrong||[]).filter(id=>db.questions.some(x=>x.id===id&&x[l])).leng
  topics:db.topics.map((t,i)=>({n:i+1,title:ttl(t,l),count:tc(l,i+1),best:q.w.u?.
app.post('/api/test/start',needUser,(q,r)=>{
  const{mode,topic}=q.body,l=lg(q.body.lang);let pool=db.questions.filter(x=>x[l]
  if(mode==='topic')pool=pool.filter(x=>x.topic===+topic);
  if(mode==='mistakes')pool=pool.filter(x=>(q.w.u?.wrong||[]).includes(x.id));
  const list=shuf(pool).slice(0,mode==='topic'?1000:40);
  if(!list.length)return r.status(400).json({error:l==='ru'?('mode==='mistakes'?'Ж
  const id=rid(),items=list.map(x=>({id:x.id,order:shuf(x[l].options.map((_,k)=>k
  attempts[id]={uid:q.w.u?.id,mode,l,topic:mode==='topic'?+topic:0,items};
  const side=(x,o,k)=>x[k]&&{text:x[k].text,options:o.map(j=>x[k].options[j])};
  r.json({id,mode,topic:mode==='topic'?+topic:0,total:list.length,pass:mode==='ra
    questions:items.map(it=>{const x=list.find(y=>y.id===it.id);return{kk:side(x,
});
app.post('/api/test/:id/answer',needUser,(q,r)=>{
  const a=attempts[q.params.id],[i,choice]=q.body;
  if(!a||a.uid!==q.w.u?.id||!a.items[i]||a.items[i].pick!==null||!(choice in a.it
  const it=a.items[i],x=db.questions.find(x=>x.id===it.id);if(!x)return r.sendSta
  it.pick=it.order[choice];const ok=it.pick===x.correct;it.ok=ok;
  if(q.w.u){const w=q.w.u.wrong=q.w.u.wrong||[],has=w.includes(x.id);
    if(!ok&&!has)w.push(x.id);if(ok&&has&&a.mode==='mistakes')w.splice(w.indexOf(
  r.json({ok,correct:it.order.indexOf(x.correct),explanation:{kk:x.kk?.explanatio
});
app.post('/api/test/:id/finish',needUser,(q,r)=>{
  const a=attempts[q.params.id];if(!a||a.uid!==q.w.u?.id)return r.sendStatus(400)
  const total=a.items.length,score=a.items.filter(it=>it.ok).length;
  if(q.w.u&&a.topic){const b=q.w.u.best=q.w.u.best||{},pc=Math.round(score/total*
  r.json({score,total});
});
function parseCSV(t){

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const L=t.replace(/^\\uFEFF/, '').split(/\\r?\\n/).filter(l=>l.trim()),d=(L[0].matc
return L.map(l=>{const c=[];let s='',inq=false;
  for(let i=0;i<l.length;i++){const ch=l[i];
    if(ch==='') {if(inq&&l[i+1]==='') {s+=' ';i++}else inq=!inq}else if(ch===d&
    c.push(s);return c.map(x=>x.trim())});
}
function parseQ(txt) {
  if(txt.startsWith('['))return JSON.parse(txt).map(o=>({topic:+o.topic,correct:+
  const side=(c,o)=>{const op=c.slice(o+1,o+5).filter(Boolean);return c[o]&&op.le
  return parseCSV(txt).filter((c,i)=>i||!isNaN(+c[0])).map(c=>({topic:+c[0],corre
}
app.get('/api/admin/qinfo',needAdmin,(q,r)=>r.json({review:db.questions.filter(x=
  topics:db.topics.map((title,i)=>({n:i+1,title, kk:tc('kk',i+1),ru:tc('ru',i+1)}))
app.post('/api/admin/topics',needAdmin,(q,r)=>{
  const t=(q.body.titles||[]).map(s=>String(s).trim()).filter(Boolean).slice(0,31)
  db.topics=Array.from({length:31},(_,i)=>t[i]||`$${i+1}-тақырып`);save();r.json({
});
app.post('/api/admin/questions/import',needAdmin,(q,r)=>{
  let rows;try{rows=parseQ(String(q.body.text||'').trim())}catch(e){return r.stat
  let ok=0,bad=0;
  for(const x of rows){
    const sides=['kk','ru'].filter(l=>x[l]&&x[l].text&&Array.isArray(x[l].options
    if(!(x.topic>=1&&x.topic<=31)||!sides.length||(sides.length===2&&x.kk.options
    const n={id:rid(),topic:x.topic,correct:x.correct};
    sides.forEach(l=>n[l]={text:String(x[l].text),options:x[l].options.map(String
    db.questions.push(n);ok++;
  }
  save();r.json({ok,bad});
});
app.delete('/api/admin/questions',needAdmin,(q,r)=>{
  const t=+q.query.topic;db.questions=t?db.questions.filter(x=>x.topic!==t):[];sa
});

// ===== ТАҚЫРЫП МАЗМҰНЫ =====
db.content=db.content||{};
const imgUp=multer({storage:multer.diskStorage({destination:UP,filename:(q,f,cb)=
const okN=n=>n>=1&&n<=31,C=n=>db.content[n]=db.content[n]||{videoId:null,text:{kk
app.get('/api/topic/:n',needUser,(q,r)=>{
  const n=+q.params.n,l=lg(q.query.lang),o=l==='kk'?'ru':'kk';if(!okN(n))return r
  const c=db.content[n]||{videoId:null,text:{},images:[],v=db.videos.find(v=>v.i
  r.json({n,title:ttl(db.topics[n-1],l),videoId:v?v.id:null,text:c.text[l]||c.tex
  images:c.images.map(i=>({id:i.id,caption:i[k(l)]||i[k(o)]||''})),count:tc(l,n
});
app.get('/api/topic-img/:id',needUser,(q,r)=>{
  const f=path.basename(q.params.id);if(!/^q?img-/i.test(f))return r.sendStatus(40
  r.set('Cache-Control','private, max-age=3600');r.sendFile(path.join(UP,f),e=>{i
});

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app.get('/api/admin/topic/:n',needAdmin,(q,r)=>okN(+q.params.n)?r.json(db.content
app.post('/api/admin/topic/:n',needAdmin,(q,r)=>{
  const n+=q.params.n;if(!okN(n))return r.sendStatus(404);
  const c=C(n);c.videoId=q.body.videoId||null;c.text={kk:String(q.body.kk||''),ru
});
app.post('/api/admin/topic/:n/images',needAdmin,imgUp.single('image'),(q,r)=>{
  const n+=q.params.n;if(!okN(n)||!q.file)return r.status(400).json({error:'Cyper
  C(n).images.push({id:q.file.filename,ck:String(q.body.ck||''),cr:String(q.body.
});
app.delete('/api/admin/topic/:n/images/:id',needAdmin,(q,r)=>{
  const n+=q.params.n;if(!okN(n))return r.sendStatus(404);
  C(n).images=C(n).images.filter(i=>i.id!==q.params.id);fs.rmSync(path.join(UP,pa
});
// ===== WORD/PDF ИМПОРТ ЖӘНЕ СҰПАҚ РЕДАКТОРЫ =====
const mammoth=require('mammoth'),docUp=multer({storage:multer.memoryStorage(),lim
const qimgUp=multer({storage:multer.diskStorage({destination:UP,filename:(q,f,cb)
const dec=s=>s.replace(/&nbsp;/g,' ').replace(/&lt;/g,'<').replace(/&gt;/g,'>').r
const OPT=/^\\(?([A-Ea-eABВГДаБвГд])\\s*[.])\\s*(.+)$/ ,NUM=/^(?:№\\s*)?\\d{1,4}\\s*[.])
const ANS=/^(?:дұрыс\\s+жауап|жауап(?:ы)?|правильный\\s+ответ|ответ|answer)\\s*[:\\-
function parseLines(lines,wrap){
  const out=[];let cur=null,pend=[];
  const flush={()=>{if(cur&&cur.q&&cur.opts.length>=2)out.push(cur);cur=null}};
  for(const L of lines){
    const t=L.text.replace(/\\s+/g,' ').trim();
    if(!t){if(L.imgs.length){if(cur&&cur.opts.length<2)cur.imgs.push(...L.imgs);e
    const am=t.match(ANS);if(am&&cur&&cur.opts.length>=2){cur.ans=am[1];continue}
    const om=t.match(OPT);
    if(om&&cur&&cur.q){let tx=om[2].trim(),mark=L.bold;if(new RegExp(MK.source).t
      cur.opts.push({label:om[1],text:tx,mark});cur.imgs.push(...L.imgs);continue
    if(cur&&!cur.opts.length&&wrap){cur.q+=' '+t;cur.imgs.push(...L.imgs);continu
    if(cur&&cur.opts.length&&wrap&&!cur.ans&&!NUM.test(t)&&(cur.opts.length<2||!//
      flush();cur={q:t.replace(NUM,''),opts:[],imgs:[...pend,...L.imgs],ans:null};p
    }
  }
  flush();return out;
}
app.post('/api/admin/import-doc',needAdmin,docUp.single('file'),async(q,r)=>{
  const topic+=q.body.topic,l=lg(q.body.lang);
  if(!okN(topic)||!q.file)return r.status(400).json({error:'Тақырып пен файлды та
  const name=(q.file.originalname||'').toLowerCase(),saved=[];let lines=[],wrap=f
  try{
    if(name.endsWith('.docx')){
      const res=await mammoth.convertToHtml({buffer:q.file.buffer},{styleMap:['u
        const ext={'image/png':'png','image/jpeg':'jpg','image/gif':'gif','image/
        const fn='qimg-'+rid()+'+'+ext;fs.writeFileSync(path.join(UP,fn),await im
        lines=res.value.split(/<\\(?:p|li|h[1-6]|tr)>|<br\\s*\\/?>/i).map(h=>{
          const txt=dec(h.replace(/<[^>]+>/g,'')).trim(),b=[...h.matchAll(/<(strong
          return{text:txt,bold:txt.length>0&&b.length/txt.length>0.6,imgs:[...h.mat

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    }else if(name.endsWith('.pdf')){
        const d=await require('pdf-parse/lib/pdf-parse.js')(q.file.buffer);
        lines=d.text.split(/\r?\n/).map(t=>({text:t,bold:false,imgs:[]}));wrap=true
    }else return r.status(400).json({error:'Тек .docx немесе .pdf файл жүктеңіз'})
}catch(e){saved.forEach(f=>fs.rmSync(path.join(UP,f),{force:true}));return r.st
const used=new Set();let added=0,withImg=0,noCorrect=0;
for(const p of parseLines(lines,wrap)){
    const opts=p.opts.slice(0,5);if(opts.length<2)continue;
    const marks=opts.map((o,i)=>o.mark?i:-1).filter(i=>i>=0);let c=marks.length==
    if(c<0&&p.ans){const U=p.ans.toUpperCase();c=opts.findIndex(o=>o.label.toUppe
    const ok=c>=0&&c<opts.length,n={id:rid(),topic,correct:ok?c:0,review:true,[1]
    if(!ok){n.note='Дұрыс жауап анықталмады. Белгілеп, сақтаңыз';noCorrect++}
    if(p.imgs[0]){n.img=p.imgs[0];used.add(p.imgs[0]);withImg++}
    db.questions.push(n);added++;
}
saved.filter(f=>!used.has(f)).forEach(f=>fs.rmSync(path.join(UP,f),{force:true}
r.json({added,withImg,noCorrect});
});
const qOut=x=>({id:x.id,topic:x.topic,correct:x.correct,kk:x.kk||null,ru:x.ru||nu
app.get('/api/admin/questions',needAdmin,(q,r)=>{
    const t=+q.query.topic||0,s=String(q.query.q||'').toLowerCase().trim(),rv=q.que
    const a=db.questions.filter(x=>(!t||x.topic===t)&&(!rv||x.review)&&(!s||[x.kk?.
    r.json({total:a.length,page:pg,items:a.slice(pg*PS,pg*PS+PS).map(qOut)});
});
function cleanQ(b){
    const topic=+b.topic;if(!okN(topic))return{err:'Тақырыпты таңдаңыз'};const o={t
    for(const l of['kk','ru']){const s=b[l];if(!s||!String(s.text||'').trim())conti
    const op=(s.options||[]).map(v=>String(v).trim());
    if(op.length<2||op.some(v=>!v))return{err:(l==='kk'? 'Қазақша': 'Орысша')+' : жа
    o[l]={text:String(s.text).trim(),options:op,explanation:String(s.explanation|
    if(!o.kk&&!o.ru)return{err:'Кемінде бір тілде сұрақ мәтіні керек'};
    if(o.kk&&o.ru&&o.kk.options.length!==o.ru.options.length)return{err:'Екі тілдер
    const c=+b.correct;if(!(c>=0&&c<(o.kk||o.ru).options.length))return{err:'Дұрыс
    o.correct=c;return{o};
}
app.post('/api/admin/questions/approve',needAdmin,(q,r)=>{
    const t=+q.body.topic||0;let ok=0,skip=0;
    db.questions.forEach(x=>{if(x.review&&(!t||x.topic===t)){if(x.note)skip++;else{
});
app.post('/api/admin/questions',needAdmin,(q,r)=>{
    const{o,err}=cleanQ(q.body);if(err)return r.status(400).json({error:err});
    const x={id:rid(),...o,review:false};db.questions.push(x);save();r.json({id:x.i
});
app.put('/api/admin/questions/:id',needAdmin,(q,r)=>{
    const x=db.questions.find(x=>x.id===q.params.id);if(!x)return r.sendStatus(404)
    const{o,err}=cleanQ(q.body);if(err)return r.status(400).json({error:err});
    x.topic=o.topic;x.correct=o.correct;x.review=false;delete x.note;

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    ['kk','ru'].forEach(l=>{if(o[l])x[l]=o[l];else delete x[l]});save();r.json({ok:
});
app.delete('/api/admin/questions/:id',needAdmin,(q,r)=>{
    const x=db.questions.find(x=>x.id===q.params.id);if(x&&x.img)fs.rmSync(path.joi
    db.questions=db.questions.filter(y=>y!==x);save();r.json({ok:1});
});
const findQ=(q,r,n)=>{q.x=db.questions.find(x=>x.id===q.params.id);q.x?n():r.send
app.post('/api/admin/questions/:id/image',needAdmin,findQ,qimgUp.single('image'),
    if(!q.file)return r.status(400).json({error:'Сурет файлын таңдаңыз (jpg, png, w
    if(q.x.img)fs.rmSync(path.join(UP,path.basename(q.x.img)),{force:true});
    q.x.img=q.file.filename;save();r.json({img:q.x.img});
});
app.delete('/api/admin/questions/:id/image',needAdmin,findQ,(q,r)=>{
    if(q.x.img)fs.rmSync(path.join(UP,path.basename(q.x.img)),{force:true});delete
});
const PUB=fs.existsSync(path.join(__dirname,'public'))?path.join(__dirname,'publi
if(PUB)app.use(express.static(PUB));
else app.get(/^\/(?:[a-z]+\.(?:html|css|js))?$/, (q,r,n)=>{
    const f=q.path==='/'?'index.html':q.path.slice(1);
    const ok=['index.html','app.html','admin.html','questions.html','test.html','to
    ok.includes(f)?r.sendFile(path.join(__dirname,f)):n();
});
app.listen(PORT,()=>console.log(`http://localhost:${PORT}`));

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